

CS673 Project Proposal

Project Name: GradAid

Group Name: Admissions Alchemists

Github URL: <https://github.com/yuezhang0594/GradAid>

Team Profile

Yue Zhang:

Yue has a background in structural biology, with expertise in both experimental and computational methods for studying proteins. Yue is proficient in scientific programming using Python and Bash and has a strong ability to learn quickly. With a keen interest in systematic programming, Yue is eager to expand their skills in software development. In this project, Yue is focused on frontend/UI development while also contributing wherever the team needs support.

Nitin Krishna Bojji:

Nitin is a skilled professional with expertise in LLM agentic AI, currently pursuing a Master's in Computer Science. With a strong foundation in Electrical Engineering from their undergraduate studies, they bring a multidisciplinary approach to problem-solving.

A natural leader, Nitin has successfully led teams in Smart India Hackathons, demonstrating exceptional team management, strategic thinking, and innovation. Their ability to guide and collaborate makes them an asset in any high-impact project. Passionate about cutting-edge AI applications, they are driven to push the boundaries of intelligent systems and team-driven innovation.

Joseph Rissman:

Joseph has a background in biology and a passion for computer science, mathematics, and software engineering. Joseph's strengths are his attention to detail with a focus on user experience and design, organization and management skills, and his experience with presenting scientific topics to non-expert audiences. Joseph plans to build upon the foundations of database management and design he gained in his previous classes to help create the backend for this product.

Product Vision

For international students planning to attend graduate school in the US who need to draft documentation required for their applications. GradAid is an AI Driven Service Provider that provides students with freedom to be creative and tailor stories to their individual needs at a low-cost. Unlike educational consultancies, our product is available to users at any time and at a lower cost.

Proposed Project Description

Recently, the number of international graduate students in the United States from India has surpassed all other countries. According to a 2023 report, the number of Indian graduate students increased by 63 percent to 165,936 students, nearly 64,000 students more than the previous year. This presents a growing need for application assistance for prospective Indian graduate students.

Students are faced with many challenges while going through the application process for a graduate program. There are several primary requirements for each application which must be individualized to the student and the school. The first of these requirements is a Statement of Purpose (SOP). The SOP is a personal essay that serves as the student's chance to showcase their personality, motivations, and qualifications beyond their transcripts and test scores. It is a narrative that connects the student's past experiences, current aspirations, and future goals, demonstrating why they are a strong fit for a specific graduate program. The second requirement are the Letters of Recommendation (LOR), which attests to the student's skills, qualifications, and character. Additionally, students are required to create university shortlists, to focus their efforts on applications that are most likely to succeed and earn them admission to the university best suited to their strengths and goals.

Due to the difficulties that each student must overcome while applying to study in a graduate school across the globe, there is a large demand for assistance with the process. Educational consultancies are businesses in India that charge fees to students, typically in the range of \$600 to \$1000 depending on the services, that meet the demand by assisting students to generate important documents and aid in the application process. Our product "GradAid" aims to capitalize on this opportunity by leveraging large language models (LLMs) and Agents to perform the majority of the work of an educational consultant at a fraction of the cost and with additional benefits to the customer.

By leveraging LLMs and supplementary methods like Fine Tuning and Prompt engineering, GradAid will provide students SOPs and LORs within seconds at minimal cost. The student will only need to provide critical information about themselves and their goals, then GradAid will harness the power of AI to provide students with personalized documentation and application assistance. GradAid will provide students with the freedom to express their individuality by allowing them to select from different SOPs and providing options to enhance their story depending on the course and program that they intend to apply for.

GradAid offers significant advantages over traditional educational consultants beyond cost and time savings. Since GradAid is a web application, it is always available to the customer for any last-minute revisions. This allows students to access and update their SOPs and LORs at the times most convenient for them. GradAid can also draw from its extensive knowledge of each field of study, providing a significant benefit over the alternatives. For example, an educational consultant who completed their education in Business Administration would not have the specific knowledge required to understand the nuances of writing and SOP for a student applying for an Electrical Engineering program.

GradAid will be measured by many metrics to ensure the product is providing value to the customer. The most important metric is admission rate, which should improve for our customers compared to those that use educational consultants. There are several important metrics that will monitor how happy the customer is with the service provided. These include the user satisfaction score, retention rate, and referral rate. Internal metrics are also important, including processing time, token cost, maximum concurrent users, and document accuracy. Some potential metrics to measure the business values are scalability, customizability, market potential, subscription and add-on opportunities, and partnerships.

The team will design and implement the product in 3 months. The major focus of development is creating the minimum viable product (MVP) that can be used by a student to generate an SOP and LOR that will aid them in applying to graduate school. This requires that the customer can interact with the product by providing details about themselves and the school they would like to attend, then the product will generate an SOP and LOR that the student can immediately download. Additional features such as document storage, modification requests, and speech recognition will be implemented later in development if there is sufficient bandwidth. The team aims for the product to be launched by April 7th, so that there is sufficient time to gather user feedback on the finished product and collect data on user metrics. The steps necessary to achieve these goals are outlined in detail below.

Phase 1: Research & Requirement Gathering (Week 1: Feb 5 - Feb 11)

Finalize project scope (MVP features only).

Collect SOP/LOR samples across disciplines.

Define key success metrics and business strategy.

Research available LLMs (fine-tuned models vs. API-based).

Draft system architecture and user flow.

Parallel Task: Start designing UI wireframes.

Phase 2: AI Model Development & System Architecture (Weeks 2-3: Feb 12 - Feb 25)

Choose LLM (open-source fine-tuned vs. GPT-4 API-based).

Implement initial prompt engineering for SOP/LOR generation.

Design the database schema and backend architecture.

Develop an API for LLM document generation.

Start UI/UX development (interactive form for user input).

Parallel Task: Setup cloud infrastructure (AWS/GCP) for hosting.

Phase 3: Frontend & Backend Development (Weeks 4-5: Feb 26 - Mar 10)

Implement user authentication, data storage, and document generation workflows.

Develop frontend interface (React/Next.js) with interactive form.

Connect frontend with backend and LLM API.

Enable SOP/LOR preview, download options.

Optimize AI responses using feedback.

Parallel Task: Develop a basic university shortlisting module.

Phase 4: Testing & Refinement (Weeks 6-7: Mar 11 - Mar 24)

Run extensive tests for SOP/LOR quality, coherence, and relevance.

Collect feedback from real users and improve prompt tuning.

Perform UI/UX refinements for better user experience.

Fix backend bugs, improve API response time. Parallel Task: Finalize university shortlisting feature and integrate it.

Phase 5: Deployment & Marketing (Weeks 8-9: Mar 25 - Apr 7)

Deploy the platform on production servers.

Set up monitoring tools (usage analytics, AI response tracking).

Conduct a small beta launch for early adopters.

Execute outreach via LinkedIn, student forums, and university groups.

Gather final user feedback and make last-minute improvements.

Final Deliverable by Apr 7, 2025

Fully functional platform with SOP/LOR generation and university shortlisting.

Hosted online with a clean, interactive UI.

Optimized AI model providing high-quality, customizable outputs.

User authentication, data storage, and basic analytics for monitoring.

To accomplish our goals, we plan to leverage the strengths of each team member. Nitin has the most expertise with machine learning, LLM, and AI agents so he will lead the effort to design and implement the part of the product that the user will interact with to create the documents. Yue has an interest in UI design so he will lead the efforts to create the frontend of the product. Joseph has an interest in database design so he will lead the efforts to create the backend of the product. Therefore, there is a team member to lead the development of each major part of the product. Together, the team has all the necessary capabilities to create the product in the specified timelines.

Functional Features

Must Have – minimal for release, required for a useful product

- Users can provide information about themselves with an interactive form.
- User information is saved in a database for use in generating documents.
- Product will produce SOP and LOR for users.
- Users can interact with the product in a web browser.
- Users can download generated documents.

Want to Have – improves on a required feature or additional functionality

- User history will be cached and saved across sessions.
- Product will generate a list of schools to apply for.
- Users can specify the format of downloaded documents.
- Document management system to update submitted information and access saved files.

Nice to Have – things to do if there is time

- Users can log in and all history will be saved across devices.
- Product website is inviting with a modern design.
- Product has speech recognition and speech generation capabilities.

References

The United States Remains the Top Choice for Indian Students Pursuing Higher Education Abroad

<https://in.usembassy.gov/the-united-states-remains-the-top-choice-for-indian-students-pursuing-higher-education-abroad/>

How Much Do Study Abroad Consultants Charge?

<https://www.aecoverseas.com/blog/abroad-consultants-charge/>